A02 - Gurukul: The Silent Instruction

Category: Reverse Engineering | **Flag:** `PINAKA{gurukul_the_silent_instruction_is_sounded}`

The challenge

We get a stripped VAC emulator, a firmware ROM, and a service that wants eight decimal "mantra" bytes. The emulator prints whether the instrument activated, the lamp output (`deepa`), and a trace tag computed over the executed instructions. The service checks both the mantra and that tag, so the input has to come from a real run of the binary, not just a guess that looks right.

Reversing the VM

The annoying part of this VM is that an opcode's meaning depends on the current mode. In the disassembly of the interpreter, the low nibble picks a register, the high nibble gets multiplied by the active mode's multiplier, and the result mod 16 picks the operation. On top of that it has four memory banks, memory-mapped bell/lamp/activation ports, and a timer interrupt that updates a seed with `x² + 29 mod 251`.

The firmware compares each of the eight bytes against a per-position constant mod 251. Each correct byte changes the later control flow and the interrupt timing, so I couldn't solve them all at once. I recovered them one at a time, replaying the real emulator after each one. The mantra is:

`[23, 86, 21, 23, 124, 206, 180, 138]`

With those bytes the binary reports `activated=1`, `deepa=2e` and trace tag `fa794a8caf8596350aaa43ba2367385cf904397c85bf52cd0e8d68f20746371e`. The same bytes do nothing on the decoy `firmware_v1.rom`, which is a good sanity check that the right ROM was used.

Getting the flag

`solve.py` runs the bundled binary with the mantra, checks the activation and trace, and submits both to the endpoint:

```
./repro.sh
python3 solve.py --submit
```

The script only replays the final answer. The work is in the VM analysis above.

```
Permissions Size User      Date Modified Name
.rw-rw-r--  1.0k darknotch 31 Oct 04:08  calibration.dat
.rw-rw-r--   904 darknotch 30 Oct 02:00  firmware.rom
.rw-rw-r--   904 darknotch 31 Oct 04:08  firmware_v1.rom
.rw-r--r--  9.3k darknotch  1 Nov 22:48  source_archive.zip
.rwxrwxr-x   19k darknotch 30 Oct 02:00  vac_emu
vac_emu:      ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=1c8ac9f44fd9de063e35a
05bd9b16daf7f0382a4, for GNU/Linux 3.2.0, stripped
firmware.rom: data

…/Pinaka/A02_player ❯
```

```
879cfa3b6bf2fab5a377c9ee14d23183094123dd67f1977028394c738c075129  firmware.rom
214f3c605a1f2e6264a7321e9fd16f2e747da1420708d333061cd8a9b955077  firmware_v1.rom
a88f9dca06834aaa767fefe69a001a95407a60b2677aa2d753bb9f9c9d9bbf4b  calibration.dat
00000000: 1007 b010 e001 2040 0330 0040 2041 0330  ...... @.0.@ A.0
00000010: 0140 2042 0330 0240 2043 0330 0340 2044  .@ B.0.@ C.0.@ D
00000020: 0330 0440 2045 0330                      .0.@ E.0

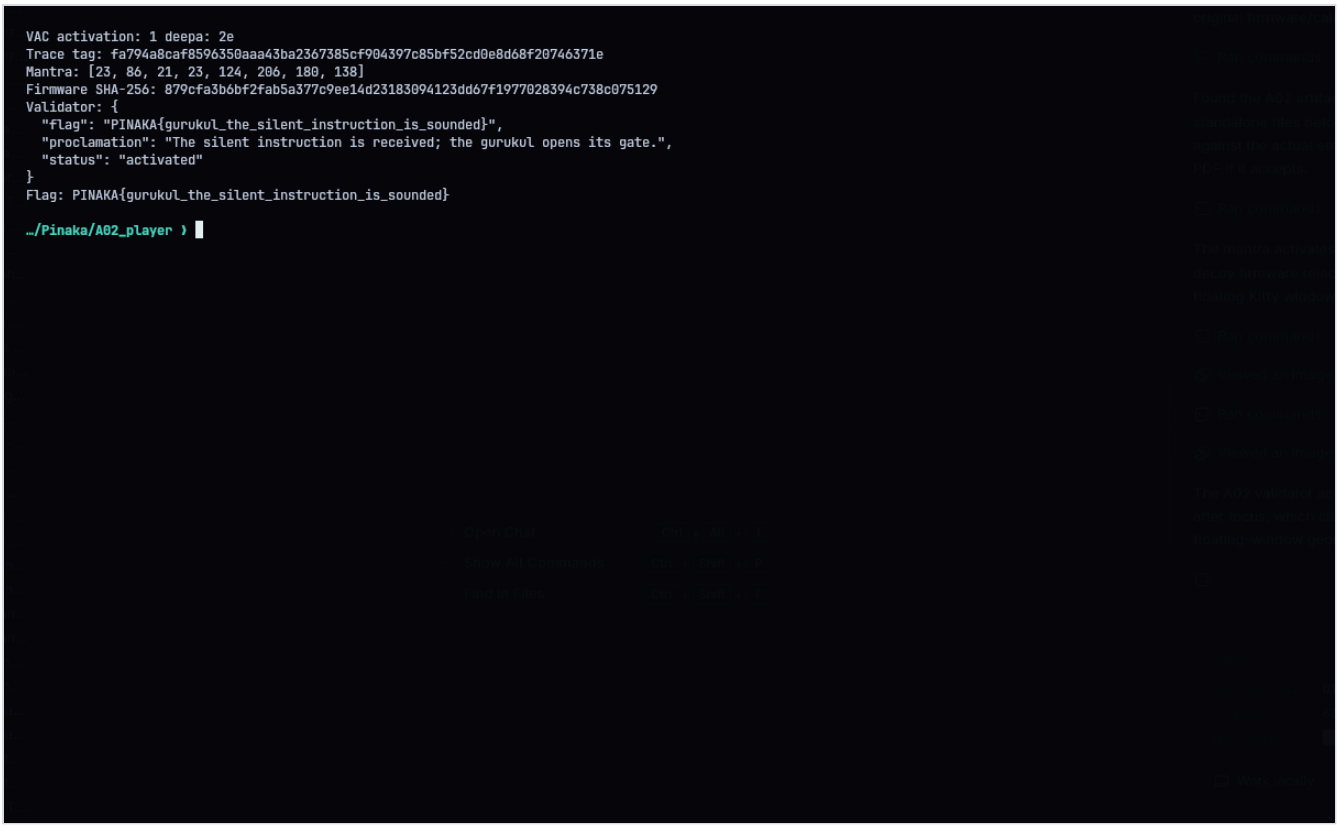
…/Pinaka/A02_player ❯
```

```
activated=1
deepa=2e
tag=fa794a8caf8596350aaa43ba2367385cf904397c85bf52cd0e8d68f20746371e

…/Pinaka/A02_player ❯
```

```
VAC activation: 1 deepa: 2e
Trace tag: fa794a8caf8596350aaa43ba2367385cf904397c85bf52cd0e8d68f20746371e
Mantra: [23, 86, 21, 23, 124, 206, 180, 138]
Firmware SHA-256: 879cfa3b6bf2fab5a377c9ee14d23183094123dd67f1977028394c738c075129

…/Pinaka/A02_player ❯
```



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